Bank Customer Churn

Prediction and Analysis

What are we solving?

- **Research Question**
  - Can we build an effective predictive model that identifies bank customers likely to churn based on historical and behavioral data?
- **Problem Statement**
  - Customer churn is a persistent and costly challenge in the financial sector. With rising competition and customer expectations, banks must proactively identify churn risks. This project aims to develop a predictive model that classifies customers into churn and non-churn categories using machine learning methods.

Why does it matter?

- **Rationale**
  - Customer churn directly impacts the revenue and sustainability of banks. Retaining customers is significantly more cost-effective than acquiring new ones. Understanding which customers are at risk of leaving allows banks to take timely and targeted actions to improve satisfaction, enhance loyalty, and reduce losses. This study enables data-driven decisions in customer relationship management.

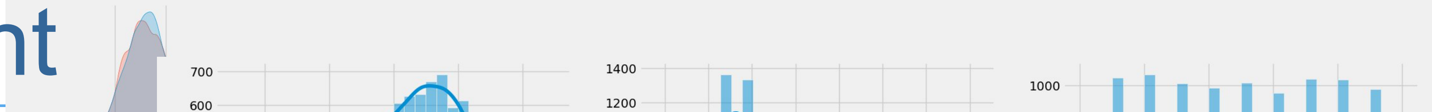
Methodology

- **Data Preprocessing**
  - Handling class imbalance by oversampling the minority class.
  - Label encoding for categorical features.
  - Outlier detection and removal using the IQR method.
  - Feature selection and engineering.
- **Model Development**
  - Logistic Regression
  - ID3, C4.5, and CART Decision Trees
  - Gaussian and Bernoulli Naive Bayes
  - Random Forest
  - AdaBoost
  - Artificial Neural Network

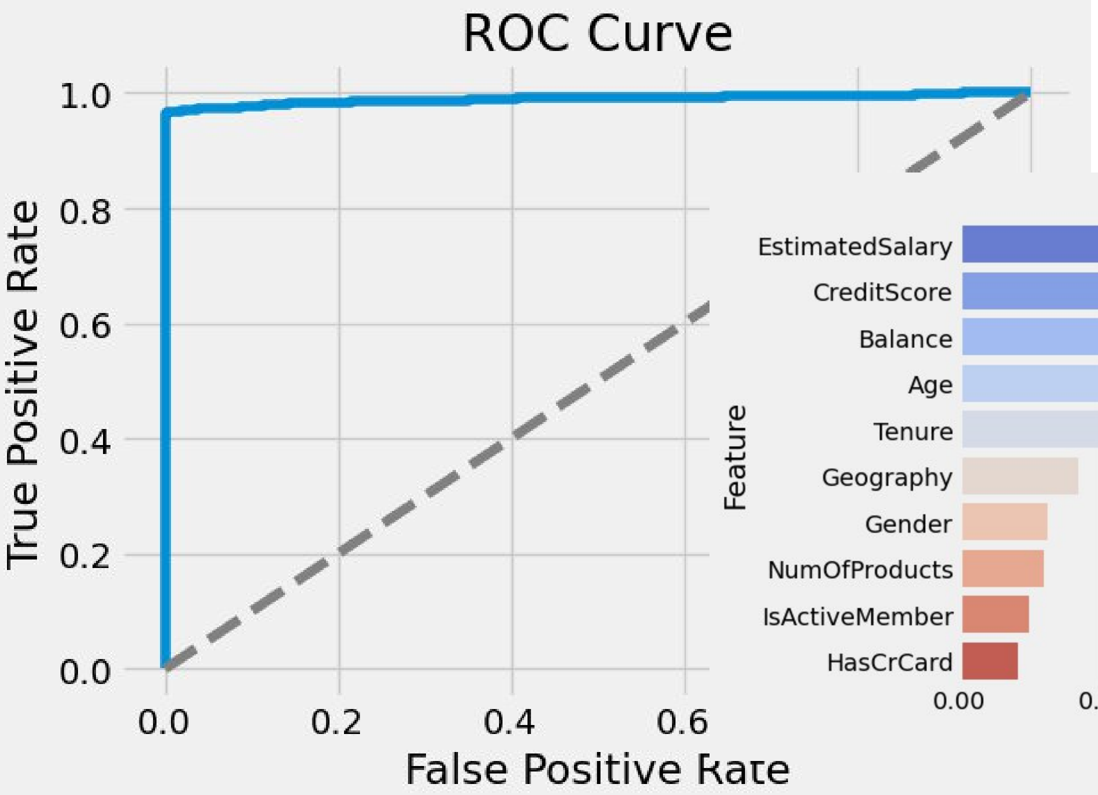
Methodology

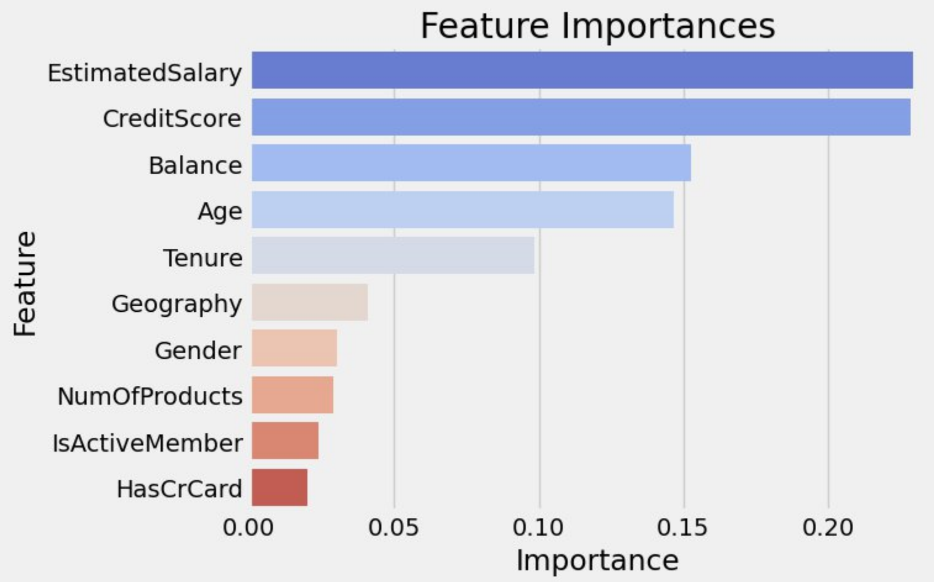
- **Evaluation**
  - Hyperparameter tuning: Grid search.
  - Cross-validation: 5-fold cross-validation.
  - Metrics: Accuracy, Cross-Validation Accuracy, Cohen's Kappa, Precision, Recall, F1-Score, Confusion Matrix, ROC-AUC and Learning Curves.

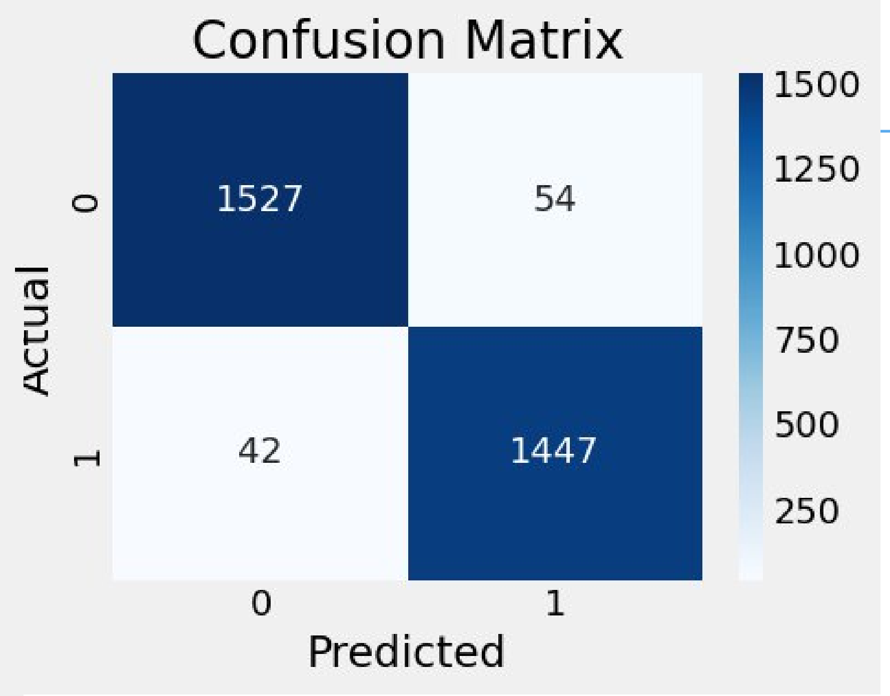
Data Preprocessing Insight

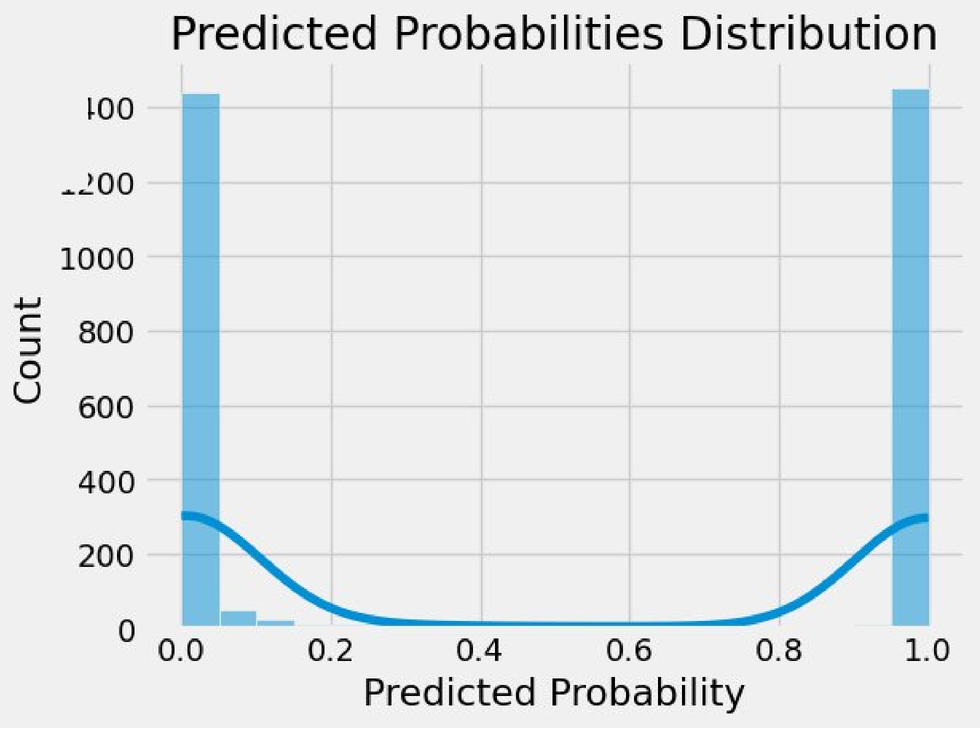


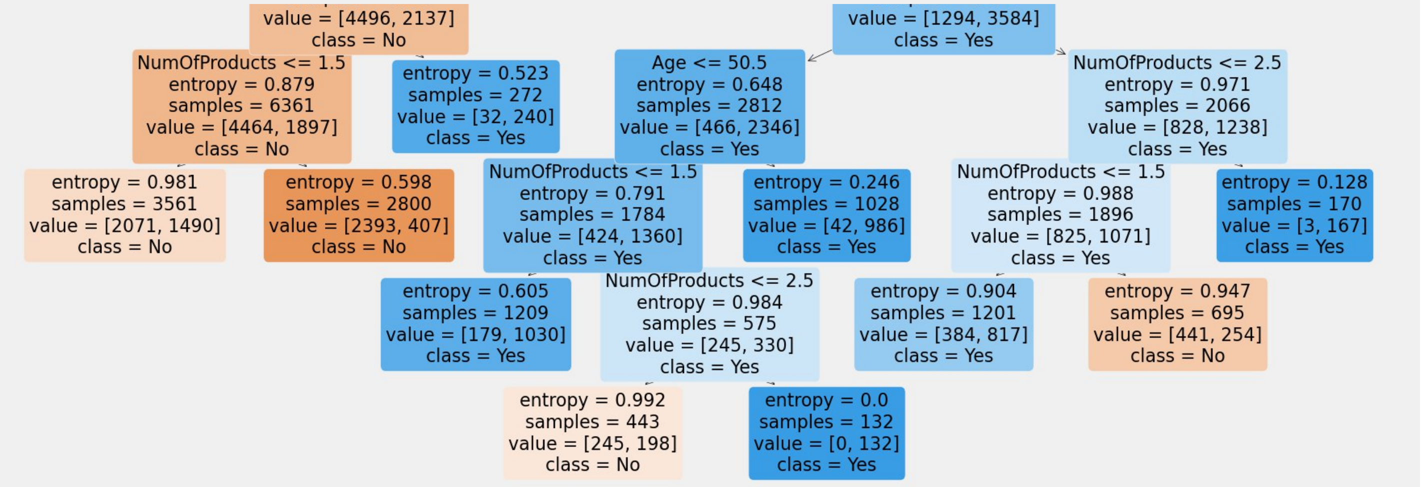
Model Prediction and Analysis Insight



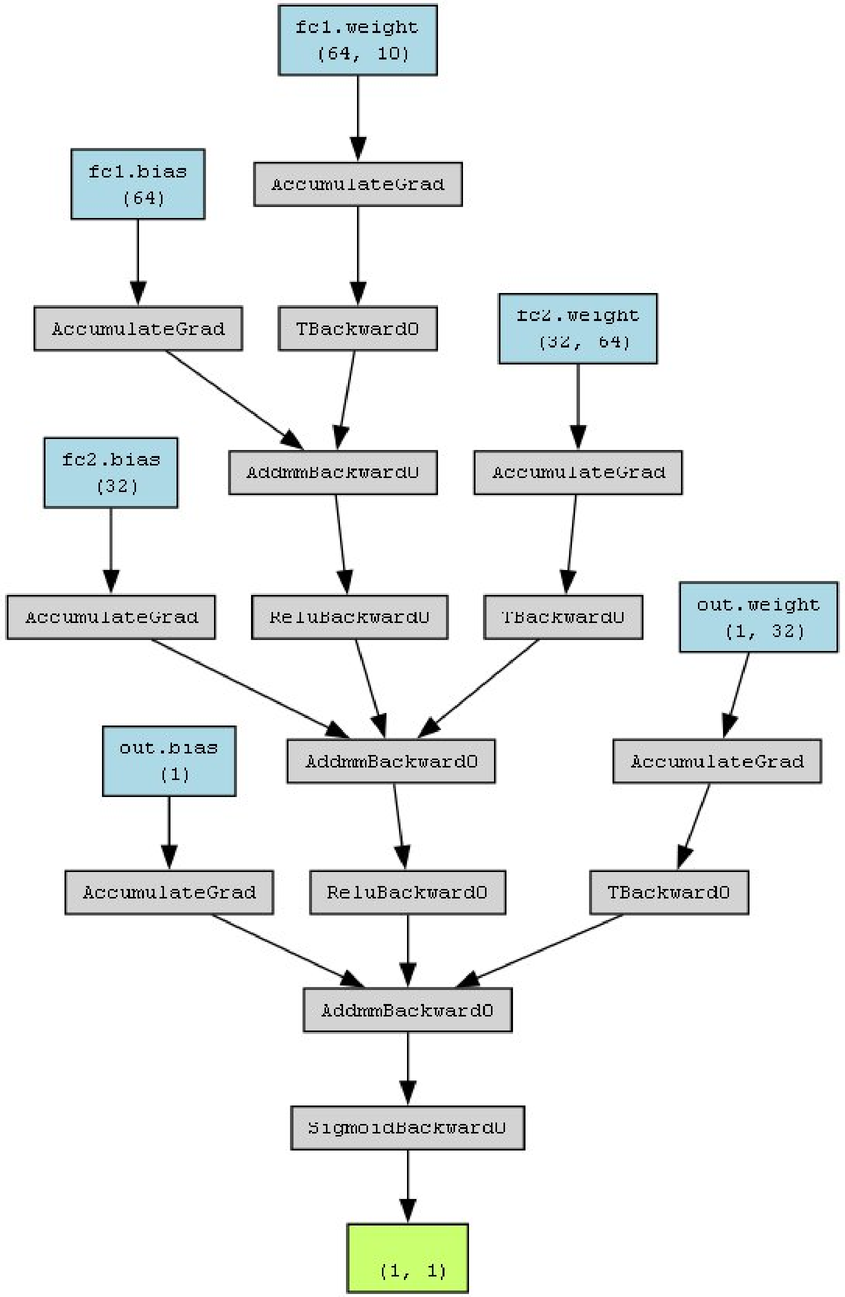


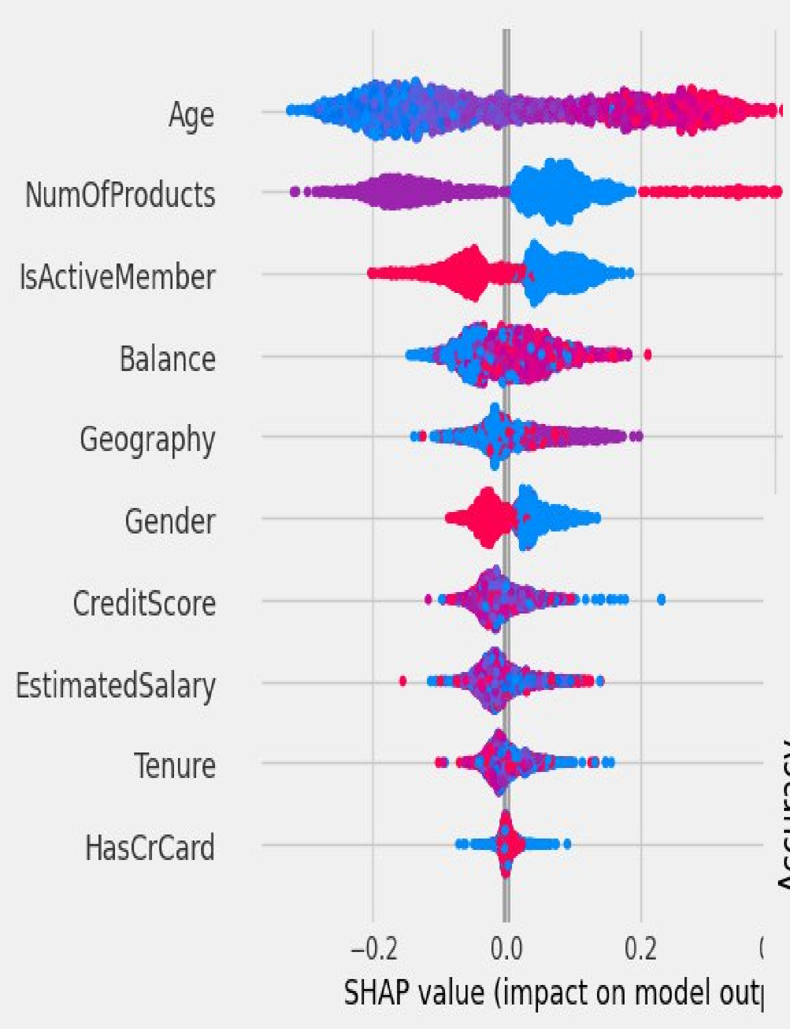


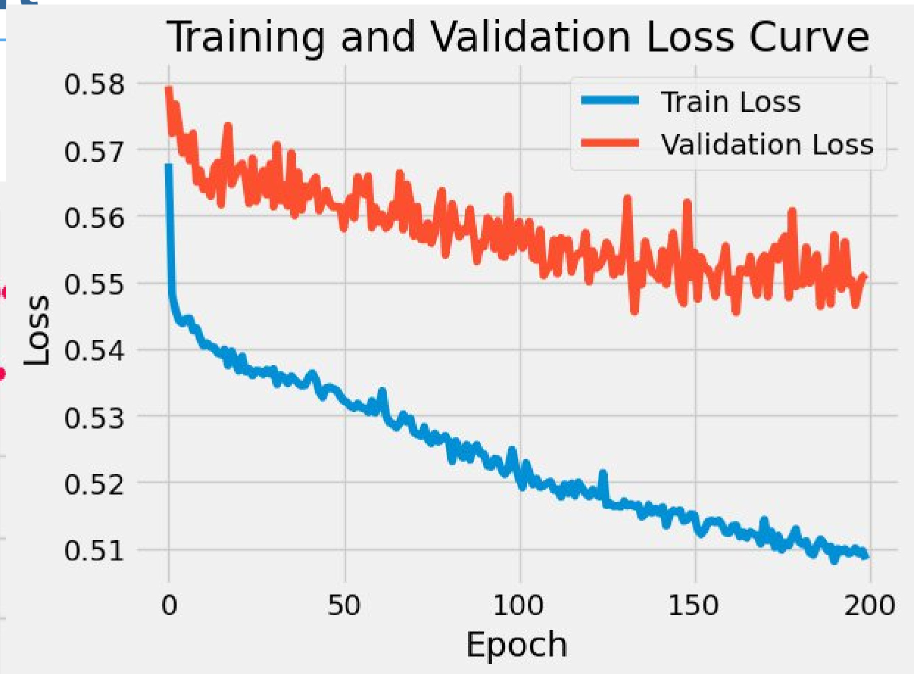


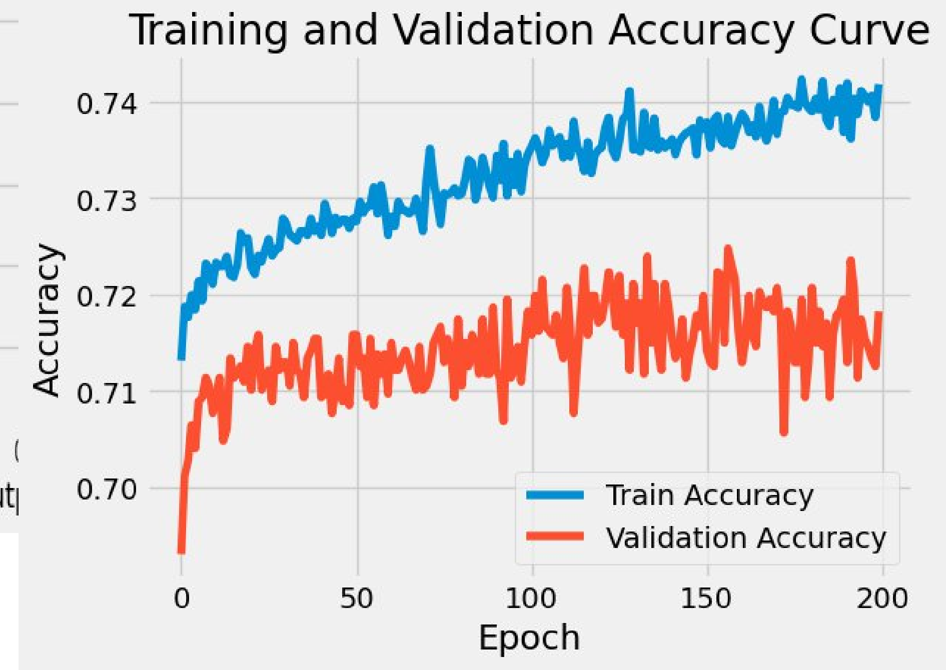


Model Prediction and Analysis Insight



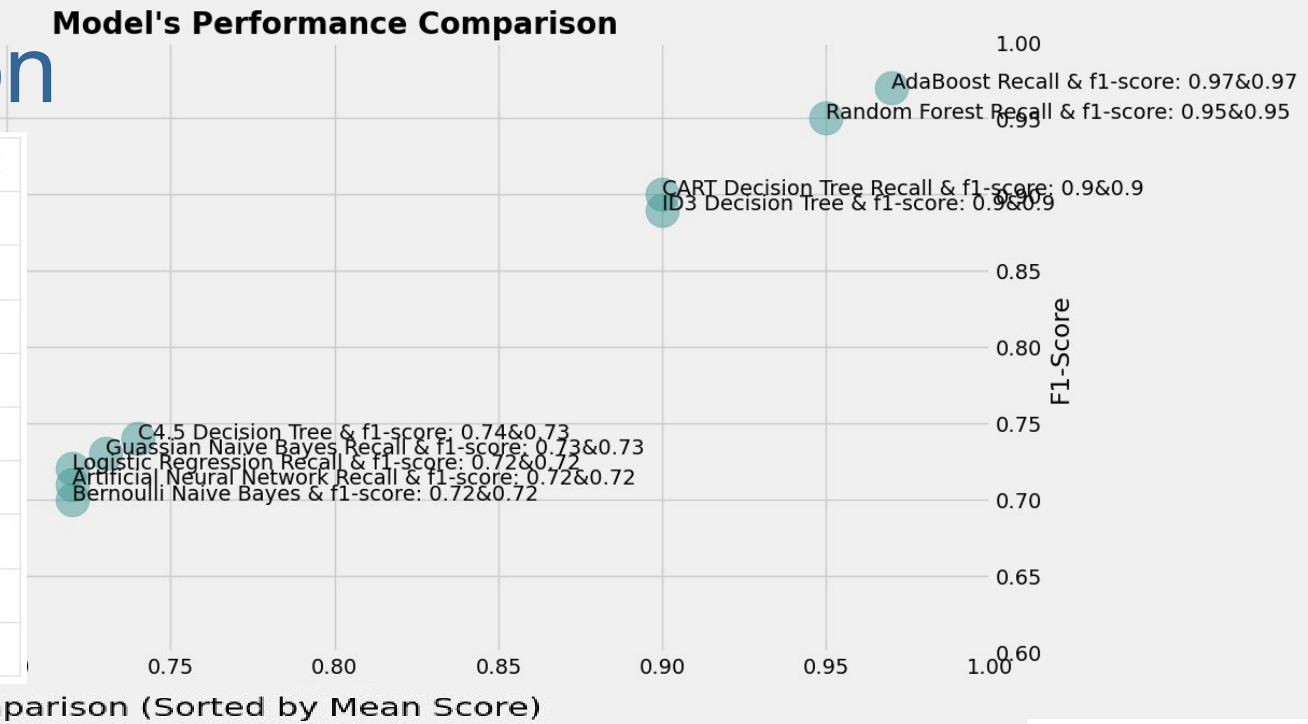


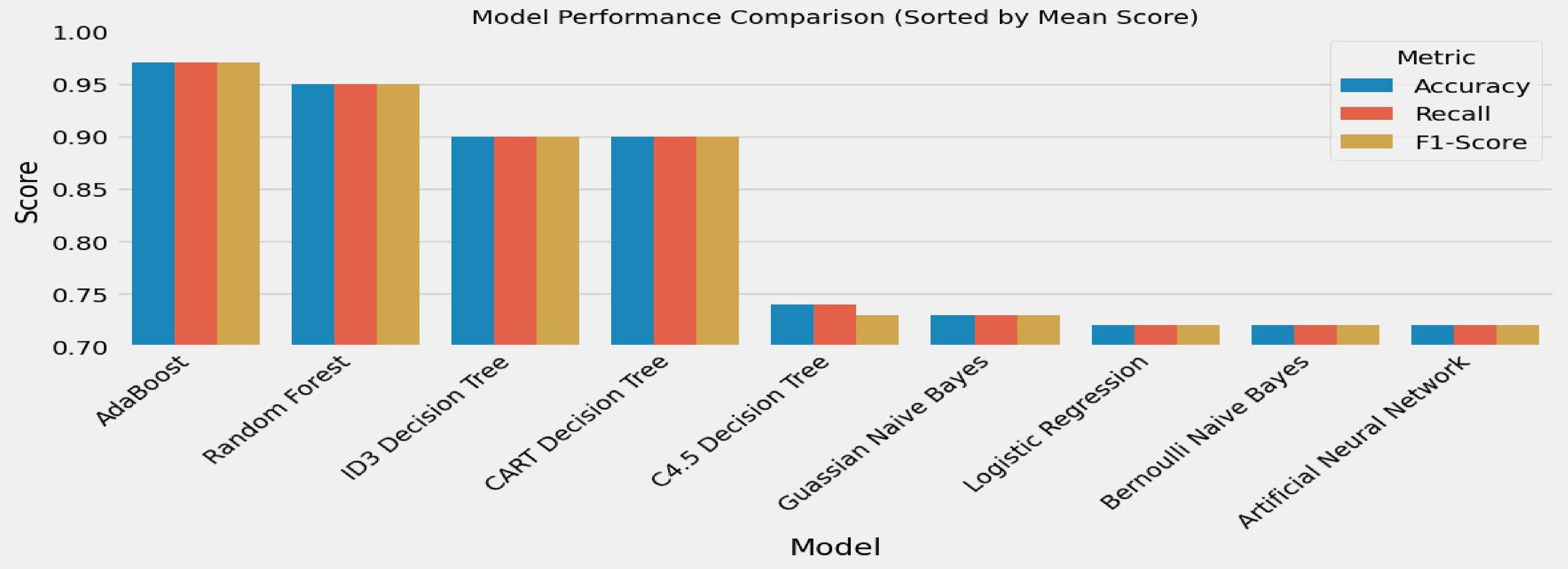




Result and Model Comparison

Model	Accuracy	CV Accuracy	Kappa	Precision	Recall	F1-Score	AUC
Logistic Regression	0.72	0.71	0.44	0.72	0.72	0.72	0.79
ID3 Decision Tree	0.90	0.89	0.80	0.91	0.90	0.90	0.90
C4.5 Decision Tree	0.74	0.73	0.47	0.76	0.74	0.73	0.81
CART Decision Tree	0.90	0.88	0.80	0.91	0.90	0.90	0.90
Gaussian Naive Bayes	0.73	0.74	0.47	0.73	0.73	0.73	0.81
Bernoulli Naive Bayes	0.72	0.72	0.44	0.72	0.72	0.72	0.78
AdaBoost	0.97	0.97	0.94	0.97	0.97	0.97	0.99
Artificial Neural Net	0.72	0.72	0.45	0.72	0.72	0.72	0.80
Random Forest	0.95	0.94	0.90	0.95	0.95	0.95	0.99





Summary

- **Conclusion**
  - This project successfully demonstrates the viability of using machine learning models to predict customer churn in the banking sector. Ensemble models like AdaBoost provide outstanding predictive performance, enabling banks to identify at-risk customers with high confidence. While the results are promising, future enhancements and real-world integration are essential for sustainable impact. Care must also be taken to ensure fairness and transparency in automated decision-making.

- **Next Steps**
  - Deploy the AdaBoost model in a real-time bank environment.
  - Automate model retraining using updated customer data.
  - Monitor performance over time and continuously refine based on feedback.